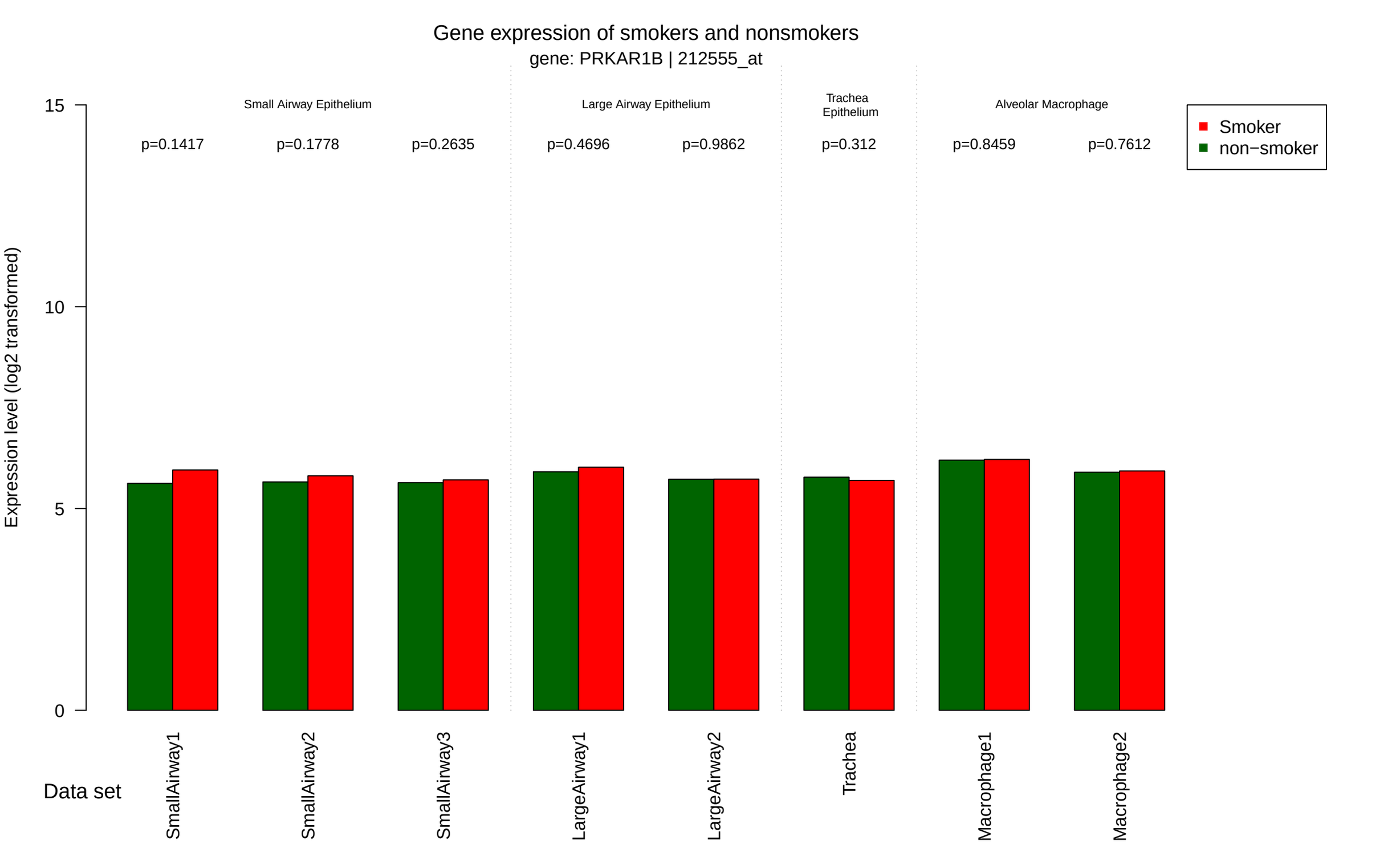
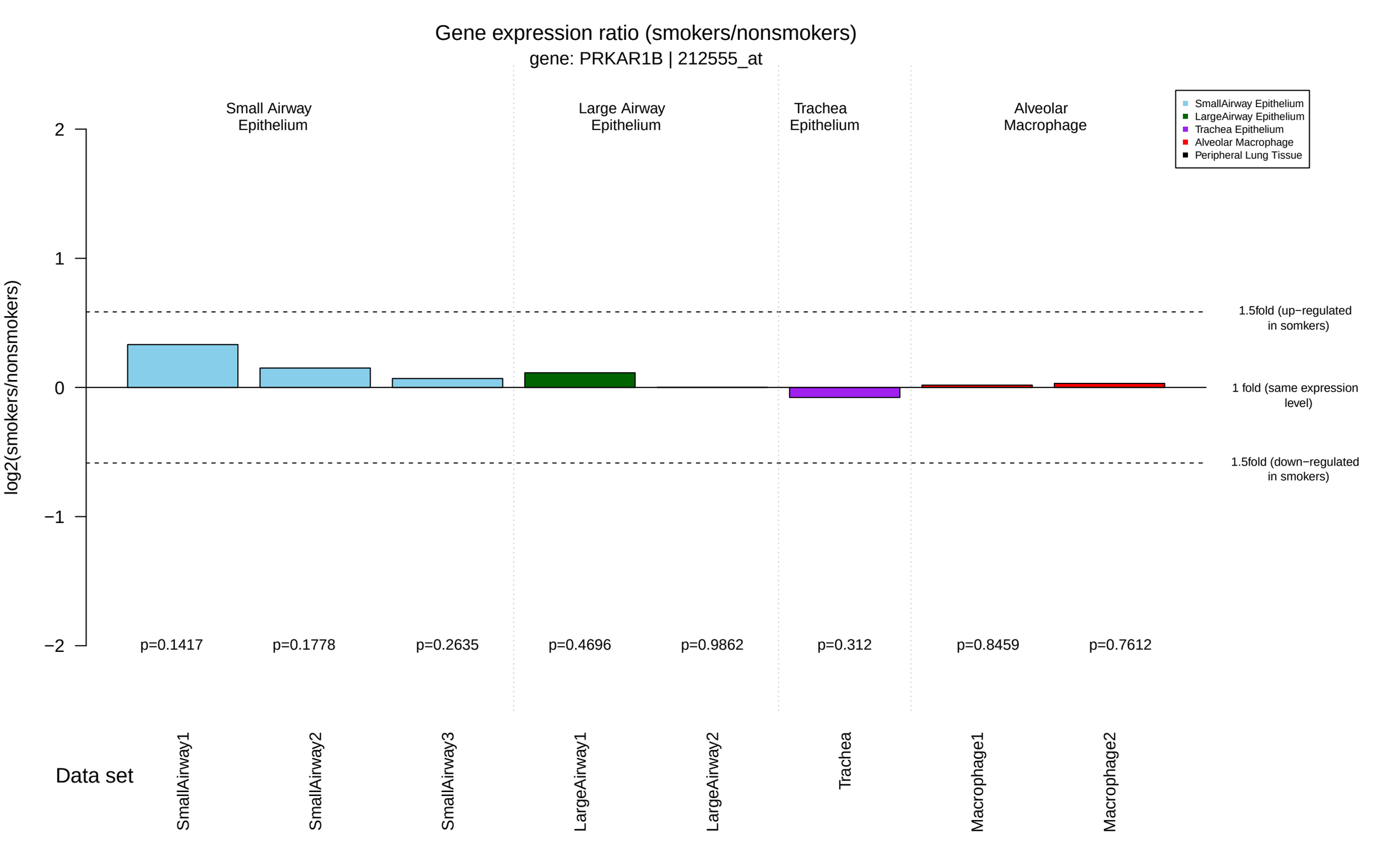
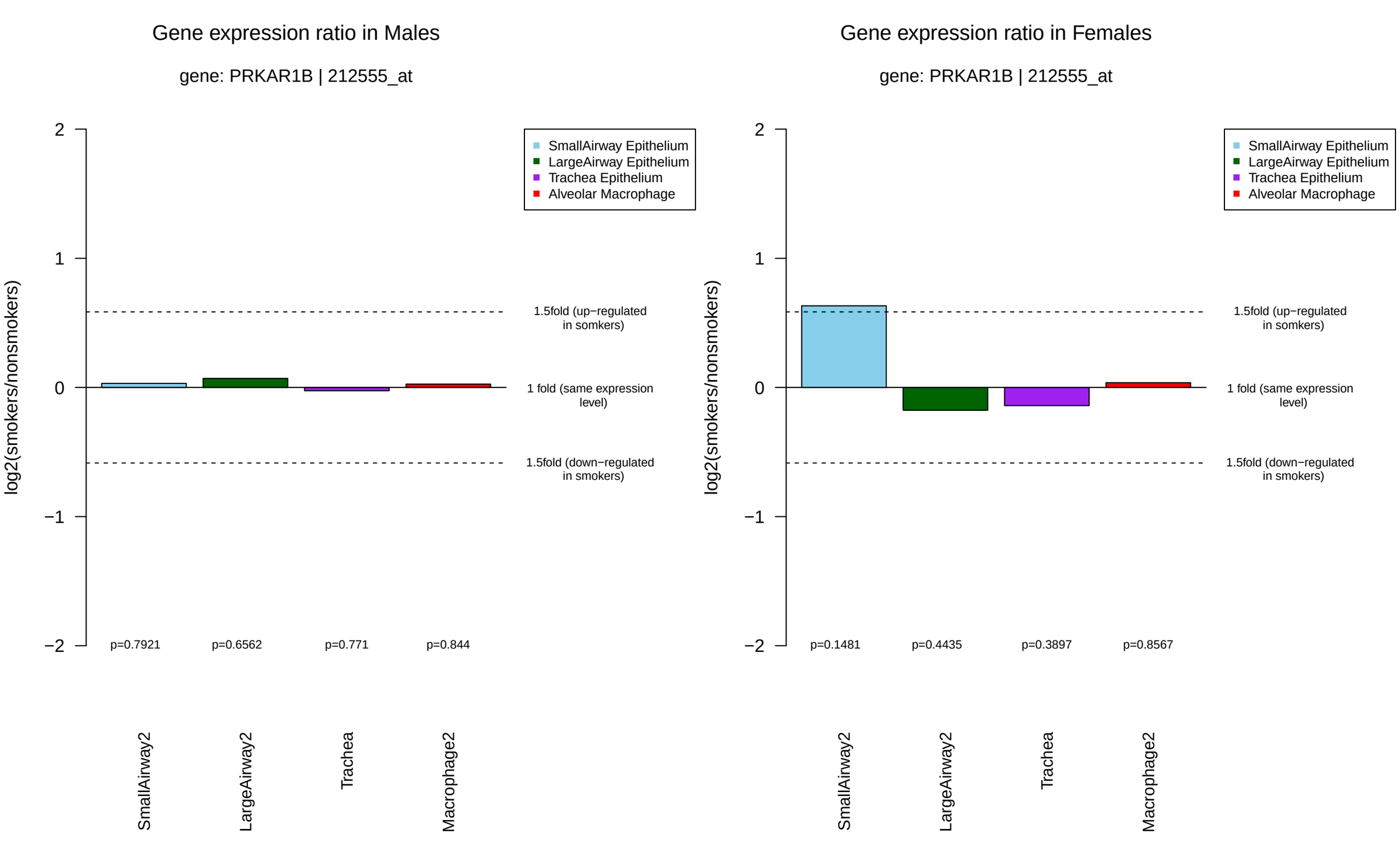
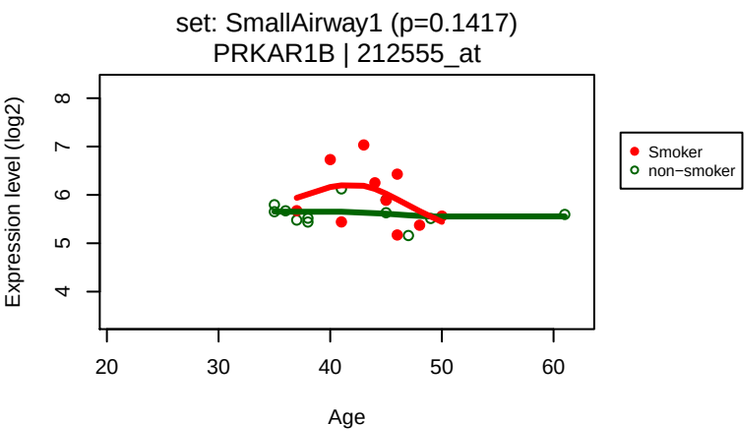
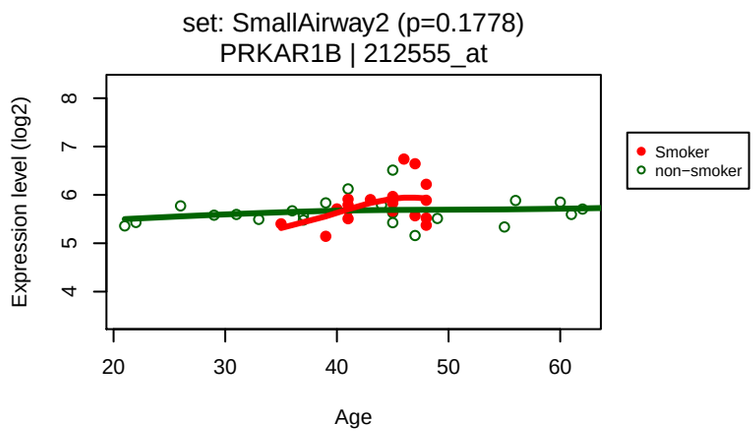
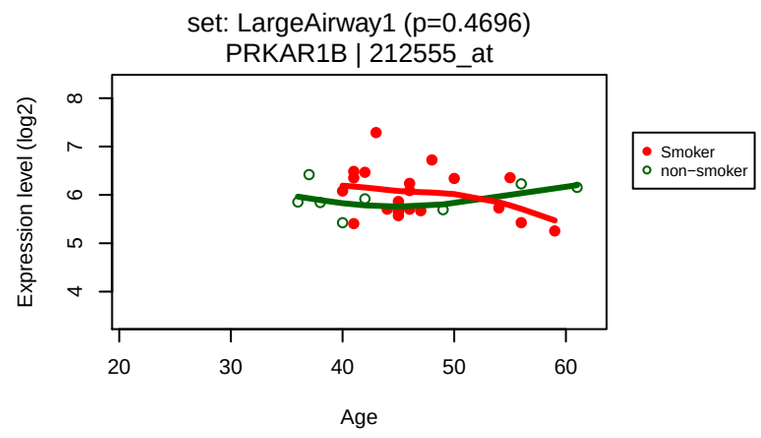

Gene expression of smokers and nonsmokers
gene: PRKAR1B | 212555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1417
p=0.1778
p=0.2635
p=0.4696
p=0.9862
p=0.312
p=0.8459
p=0.7612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

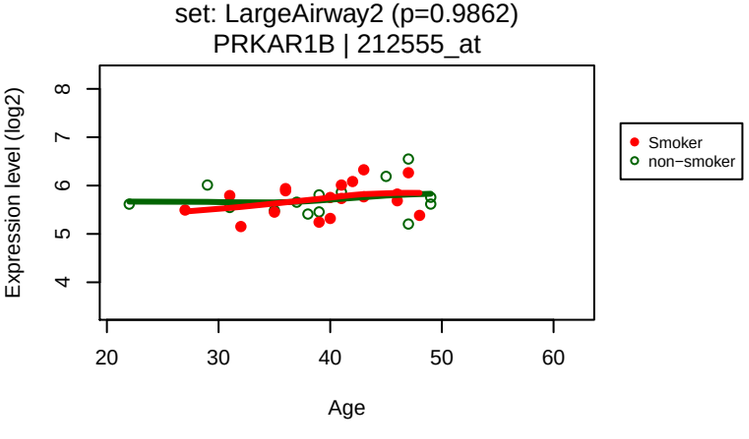

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR1B | 212555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1417
p=0.1778
p=0.2635
p=0.4696
p=0.9862
p=0.312
p=0.8459
p=0.7612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

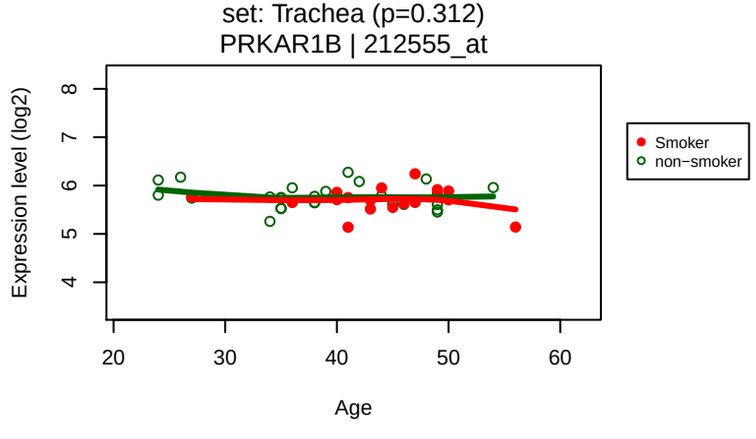

Gene expression ratio in Males
gene: PRKAR1B | 212555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7921
p=0.6562
p=0.771
p=0.844
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR1B | 212555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1481
p=0.4435
p=0.3897
p=0.8567
SmallAirway2
LargeAirway2
Trachea
Macrophage2

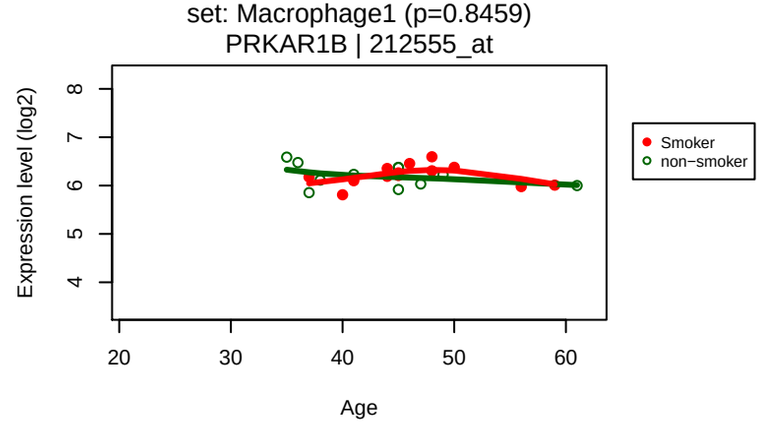

set: SmallAirway1 (p=0.1417)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker

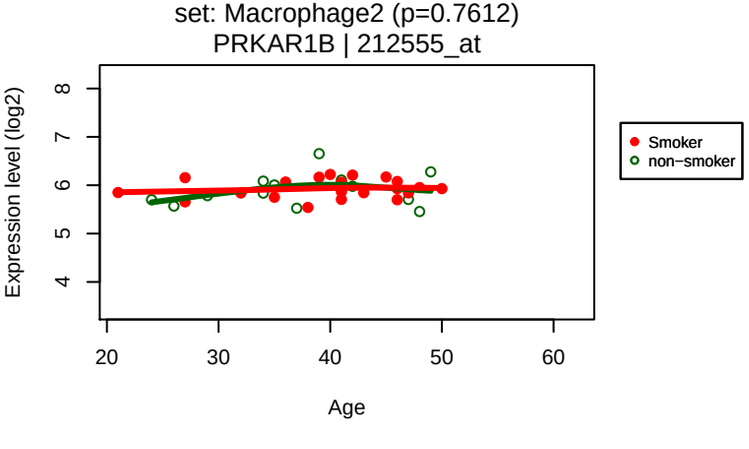

set: SmallAirway2 (p=0.1778)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker

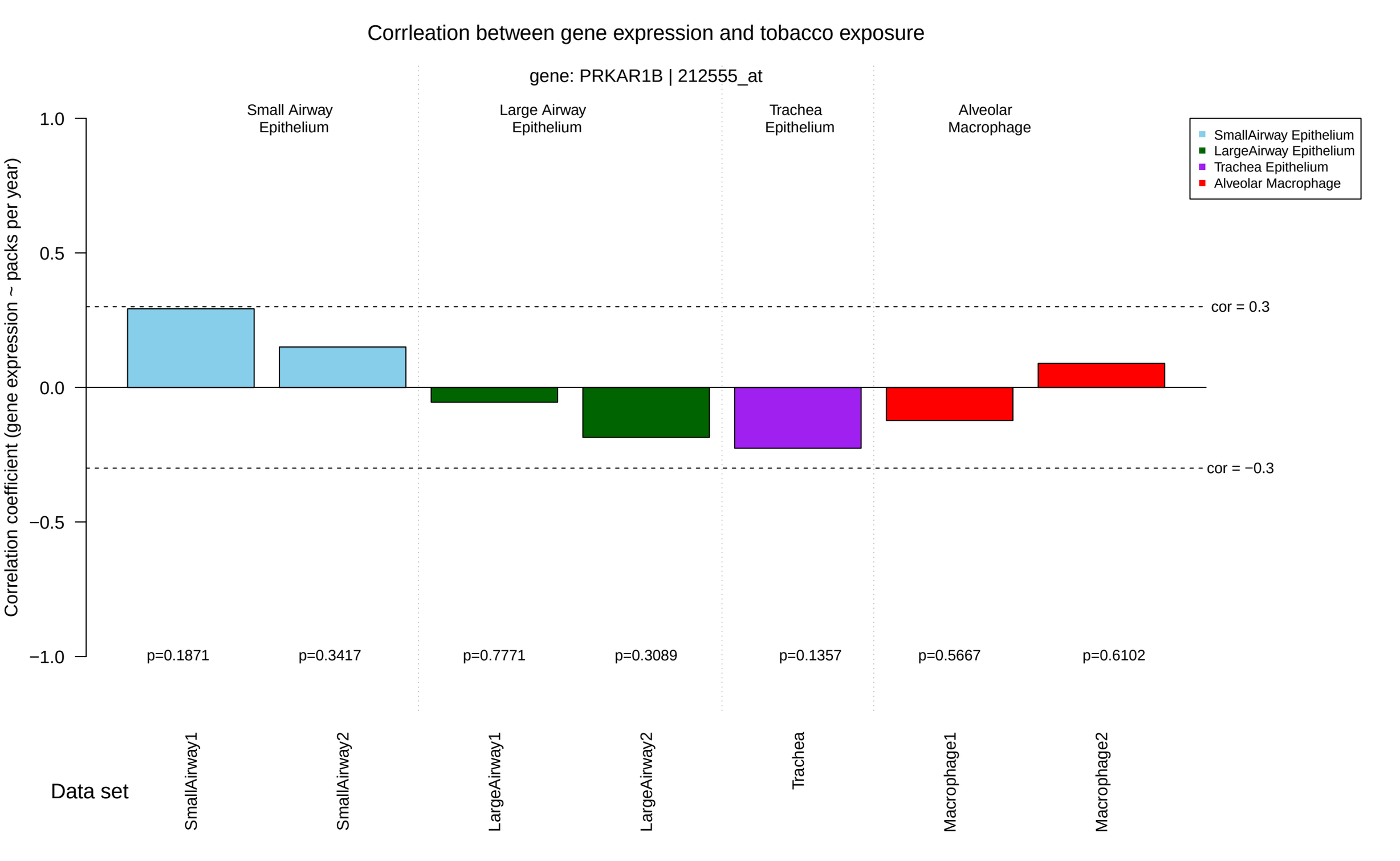

set: LargeAirway1 (p=0.4696)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9862)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.312)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8459)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7612)
PRKAR1B | 212555_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR1B | 212555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1871
p=0.3417
p=0.7771
p=0.3089
p=0.1357
p=0.5667
p=0.6102
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2